

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

September 20, 2026

Where cancers are caught, and whether anyone looks

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Target
application

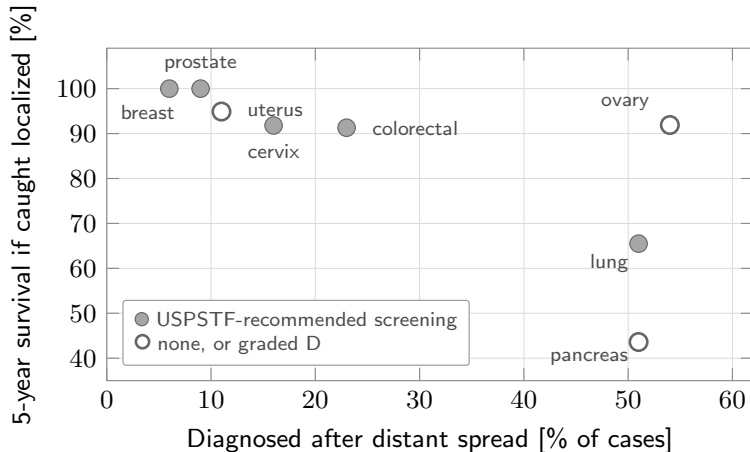
The chip

Analog front end

From currents to
a score

References

Backup



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why ovarian: late, lethal, and unscreened

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Target application

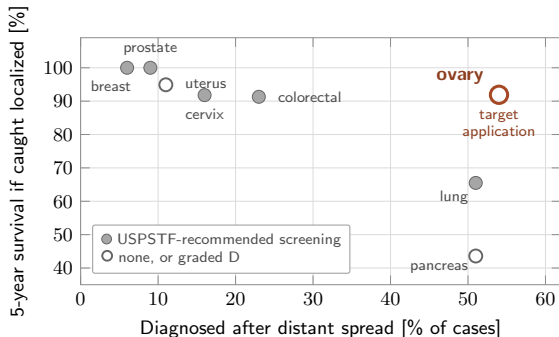
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Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

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Target application

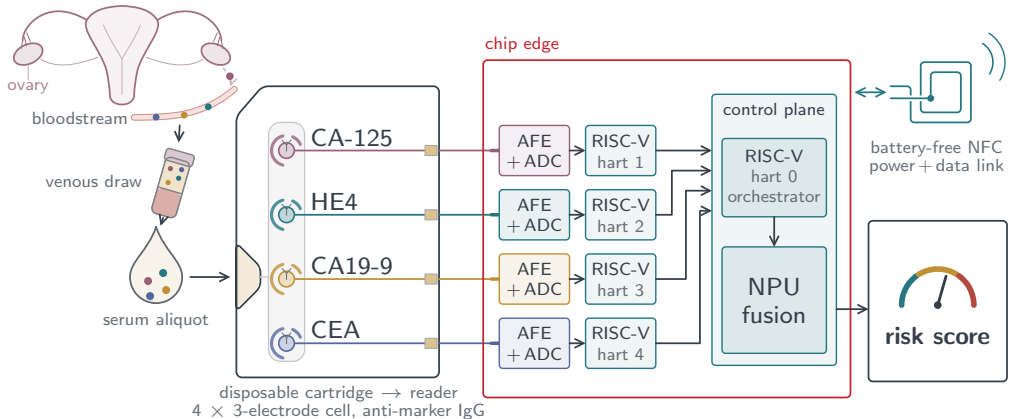
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Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single marker is a test; four fused are [5]

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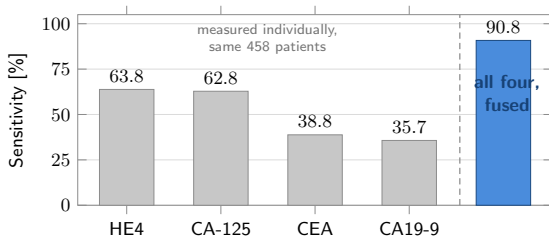
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Marker	Molecule	Cut-off
CA-125	MUC16 mucin epitope [8]	35 U/mL
HE4	WFDC2 glycoprotein [6]	70 pmol/L
CA19-9	sialyl-Lewis ^a antigen	37 U/mL
CEA	CEACAM5 glycoprotein	5 ng/mL

Each marker covers a subtype the others miss. The decision is a learned combination, so the chip is four channels plus the processor that fuses them.

Four markers resolved in one differential-pulse sweep

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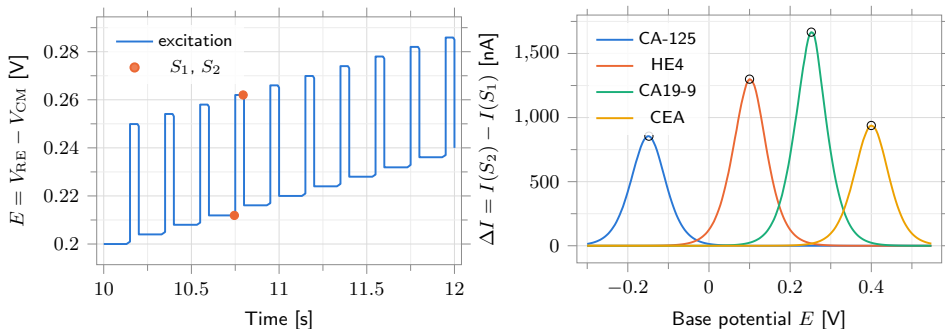
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■ schematic sim, pre-layout

Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

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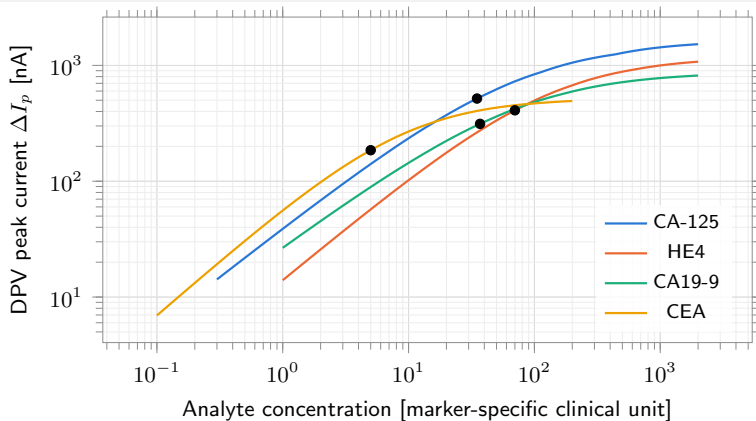
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■ schematic sim,

pre-layout

CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Castalia at a glance

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Front currents to
a scan

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Backup

Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	10.72 mW post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Block diagram

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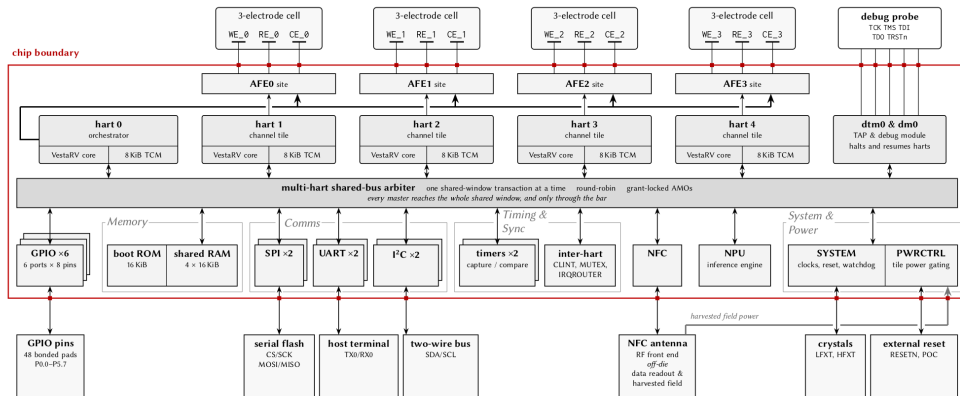
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Analogue front end

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Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

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**ARTWORK
PENDING**



chip top, 3.0×3.0 mm
(Virtuoso export)

**ARTWORK
PENDING**



one channel
tile (Virtuoso
export)

Process	65 nm, 1P8M, HVT
Core box	$2690 \times 2690 \mu\text{m}$
Tile	$660 \times 880 \mu\text{m}$
Tile logic	≈ 98 k NAND2-eq.
Tile SRAM	8 KiB
Package	LQFP-100, 14×14 mm

■ hardened flow, post-route

One tile hardened once, instantiated four times; the notch in each tile is the analog channel's keep-out.

Digital, as closed

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Whole chip, post-route

Internal	3.78 mW
Switching	2.33 mW
Leakage	4.61 mW
Total	10.72 mW

One channel tile

Total	1.761 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726
Aggregate	1 314 802	1.260

Timing at 25 MHz

reg2reg WNS	+99 ps
Leakage share	43 %

■ hardened flow, post-route

SS 0.9 V 125 °C, statistical activity 0.2. Gating one tile removes 1.76 mW; the always-on band pays nothing.

Power gating: one switched rail per channel tile

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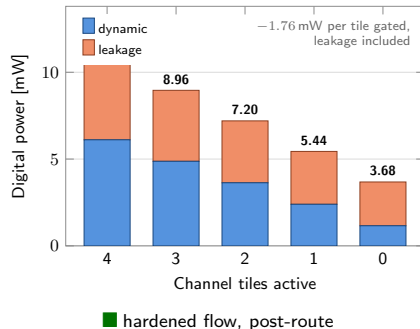
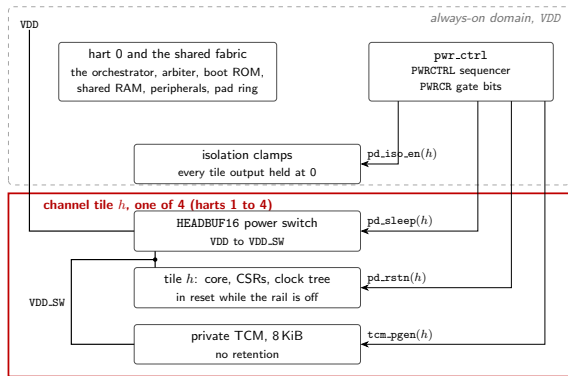
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Analog front end

From currents to
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Cold gating: the header cuts the rail, so a parked tile draws neither dynamic nor leakage power. Each tile gated removes 1.76 mW; the always-on band keeps 3.68 mW.

Grounded-WE two-amplifier potentiostat

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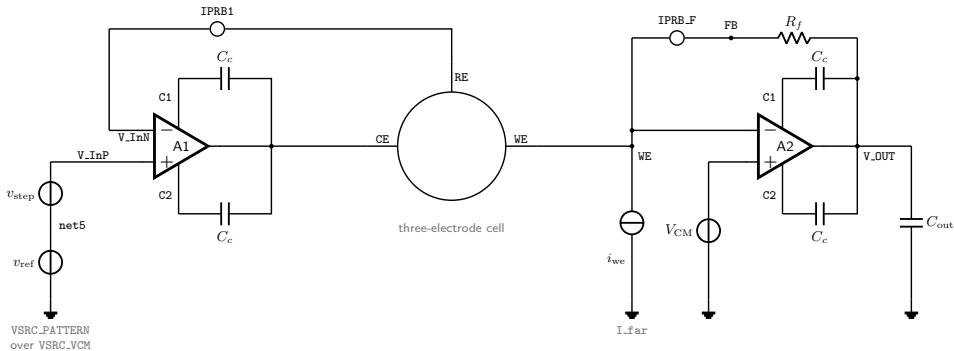
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Analog front end

From currents to
a scope

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A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we}R_f$, both polarities.

Analog top level

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ARTWORK PENDING



Virtuoso export of the analog top level: four channels, shared bias, test bus

The class-AB amplifier: one cell, both roles

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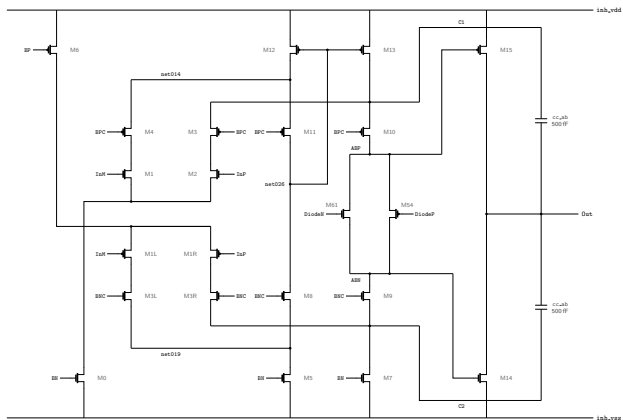
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Analog front end

From currents to
a scope

References

Backup



complementary
input pairs and
their cascodes

diode-connected
summing branch

class-AB
drive

output

testbench
compensation

A1 across corners

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From currents to a score

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Open-loop gain A_{OL}	dB	82.77	79.88	83.14	83.37	84.94	5.06
Phase margin	°	73.52	74.70	74.04	72.86	73.34	1.84
Dominant pole $f_{-3\text{dB}}$	kHz	1.592	2.286	1.538	1.494	1.225	1.061
Gain-bandwidth product	MHz	22.781	23.187	22.894	22.939	22.587	0.600
Parameter	Unit	tt	ff	fs	sf	ss	Spread
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	0.193

■ schematic sim, pre-layout

A1 under mismatch

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Target
application

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Analog front end

From currents to
a score

References

Backup

Parameter	Unit	N	Mean	σ	p_1	p_{99}	Worst	Spec	Yield	4 ch
Open-loop gain A_{OL}	dB	200	82.16	1.06	80.04	84.60	79.50	≥ 40.00	100.0	100.0
Phase margin	$^\circ$	200	73.62	1.42	69.80	76.21	68.93	≥ 40.00	100.0	100.0
Dominant pole $f_{-3\text{dB}}$	kHz	200	1.723	0.276	1.158	2.453	–	–	–	–
Gain-bandwidth product	MHz	200	22.815	3.566	16.191	31.741	–	–	–	–

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Input offset voltage	mV	200	0.583	5.201	0.583 ± 15.604	–14.130	15.709	± 1.000	12.0	0.0

■ Monte Carlo, 200 samples

A2 as transimpedance amplifier: corners and mismatch

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Analog front end

From currents to a score

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Loop gain at DC	dB	78.74	76.32	79.28	79.32	82.69	6.37
Loop crossover	kHz	9.375	9.475	9.428	9.428	9.360	0.115
Phase margin	°	76.50	80.30	76.23	76.13	72.78	7.52
Gain margin	dB	21.81	20.67	21.90	21.24	22.49	1.82

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Output offset at zero current	mV	200	-1.08	8.91	-1.08 ± 26.72	-28.46	25.28	± 5.00	48.0	5.3

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

Bipolar accuracy through zero

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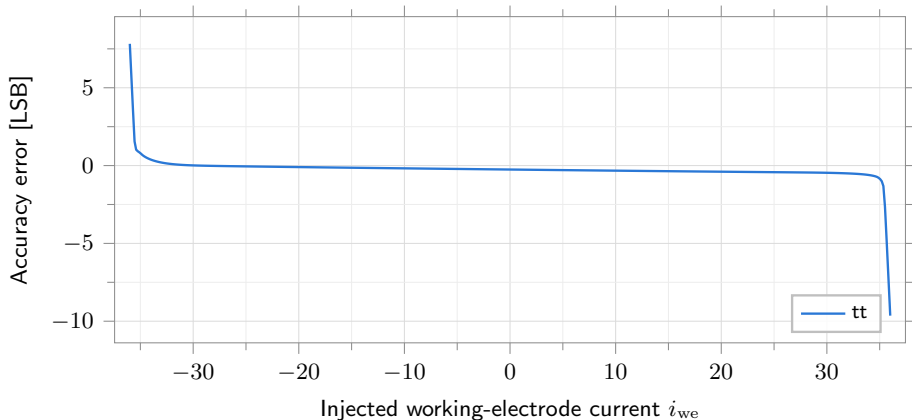
The chip

Analog front end

From currents to a score

References

Backup



■ schematic sim, pre-layout

No kink at the crossing: class-AB output, no dead zone. This is the measurement rev-1's PMOS mirror could not make.

Cyclic voltammetry through the assembled channel

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Target
application

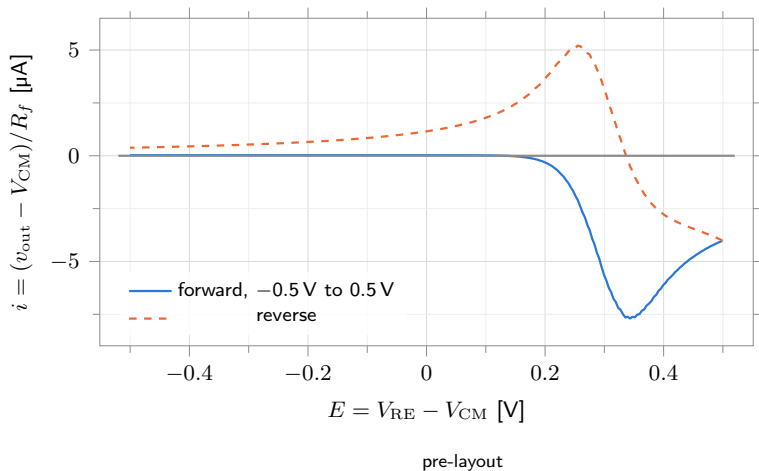
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From currents to
a scope

References

Backup



■ schematic sim,

pre-layout

Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Programmable feedback resistor: one string, 320 codes

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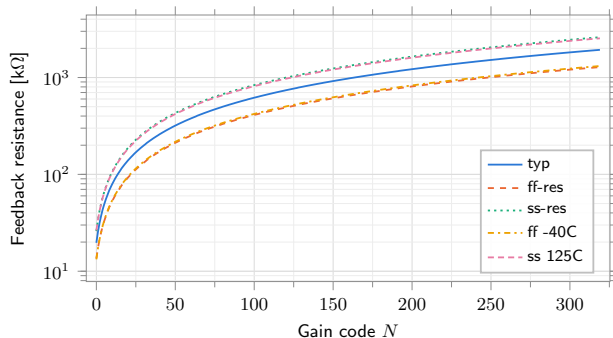
The chip

Analog front end

From currents to a score

References

Backup



$$R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$$

N	R_f	Full scale
0	19.6 kΩ	$\pm 40.7 \mu\text{A}$
63	397 kΩ	$\pm 2.02 \mu\text{A}$
319	1.93 MΩ	$\pm 0.41 \mu\text{A}$
LSB at 319		0.824 nA

Strictly monotonic at every corner, including the four thermometer seams. Maximum gain is noise-limited, not quantisation-limited.

Feedback resistor across corners and under mismatch

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Target application

The chip

Analog front end

From currents to a score

References

Backup

Parameter	Unit	typ
Chord resistance	k Ω	19.639
Worst deviation	mV	1.998
Worst deviation	ADC LSB	1.255
Secant- R modulation	%	0.504

Parameter	Unit	N	Mean	σ	Yield
$R(N=0)$	k Ω	200	19.650	0.071	100.0
$R(N=63)$	k Ω	200	396.4515	0.3608	100.0
$R(N=64)$	k Ω	200	403.8759	0.3644	100.0
$R(N=319)$	M Ω	200	1.93212	0.00071	100.0
Seam step 63 $\rightarrow$ 64	k Ω	200	7.424	0.504	100.0

■ schematic sim, pre-layout

■ Monte Carlo, 200 samples

10-bit SAR ADC, extracted netlist

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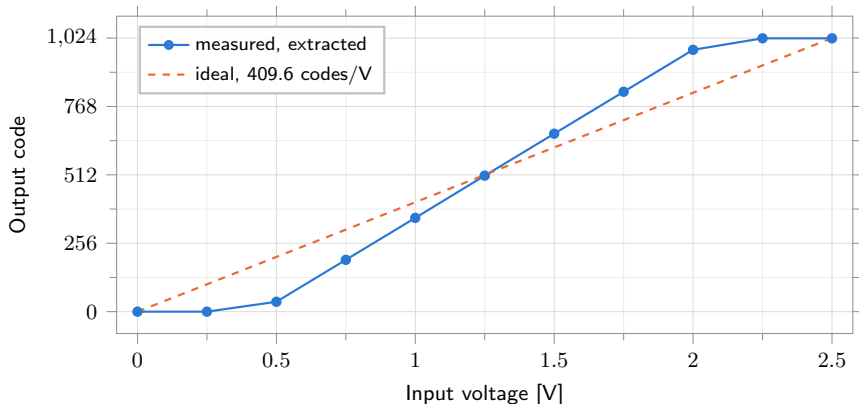
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Analog front end

From currents to
a scope

References

Backup



■ extracted

netlist

23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

SAR ADC across corners

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Analog front end

Front currents to a scope

References

Backup

Parameter	Unit	tt 37 °C	ff	fs	sf	ss
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302
Monotonic points		20/20	20/20	20/20	20/20	20/20
Conversion period	μs	1.150	1.150	1.150	1.150	1.150
Analog supply current	μA	6.360	4.787	5.453	8.437	15.429
Total power	μW	18.09	16.77	15.77	23.74	43.06
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5

■ extracted netlist

Reference DAC across corners and under mismatch

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Analog front end

Front currents to a scope

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Zero-scale output	μV	0.02	0.25	0.01	0.06	0.00	0.25
Full-scale output	V	2.499388	2.499388	2.499388	2.499388	2.499388	0.000000
LSB size	μV	610.351	610.351	610.351	610.351	610.351	0.000
INL, most positive	LSB	0.113	0.101	0.100	0.127	0.130	0.030
INL, most negative	LSB	-0.376	-0.338	-0.390	-0.362	-0.414	0.076
INL, worst magnitude	LSB	0.376	0.338	0.390	0.362	0.414	0.076
DNL, most positive	LSB	0.002	0.001	0.002	0.002	0.002	0.000
DNL, most negative	LSB	-0.423	-0.374	-0.424	-0.424	-0.479	0.105
DNL, worst magnitude	LSB	0.423	0.374	0.424	0.424	0.479	0.105

Parameter	Unit	N	Mean	σ	Median	p_1	p_{99}	Worst	Spec	Yield	4 ch
INL worst	LSB	200	1.485	0.620	1.364	0.426	3.287	3.393	≤ 1.000	20.5	0.2
INL max	LSB	200	1.154	0.610	1.031	0.194	2.886	-	-	-	-
INL min	LSB	200	-1.485	0.620	-1.364	-3.287	-0.426	-	-	-	-
DNL min	LSB	200	-1.682	1.415	-1.287	-6.161	-0.058	-6.499	≥ -1.000	42.0	3.1
DNL max	LSB	200	1.123	1.128	0.713	0.007	4.760	-	-	-	-
DNL worst	LSB	200	2.249	1.314	1.847	0.441	6.161	6.499	≤ 0.750	9.0	0.0

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

How impedance is measured: the same loop, no extra hardware

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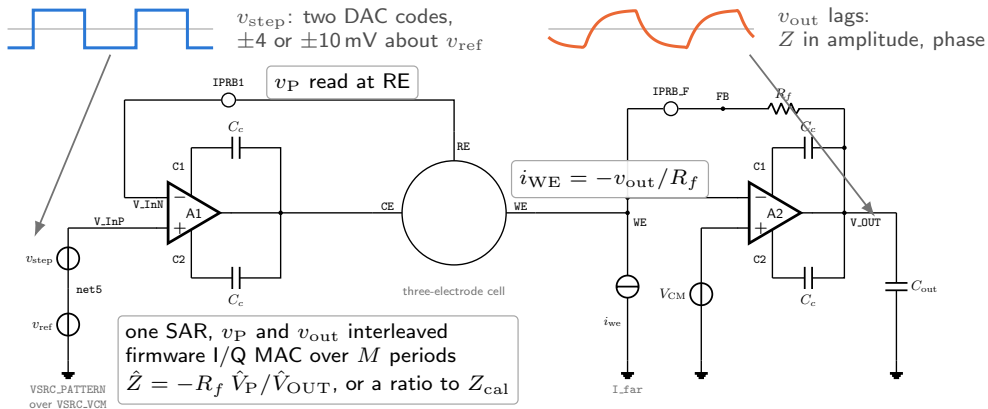
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From currents to a score

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No impedance hardware: the mode adds a clock divider and an accumulate to the channel. Against a known Z_{cal} the converter LSB, the square's $4/\pi$ and R_f all cancel.

Impedance spectroscopy on the same channel

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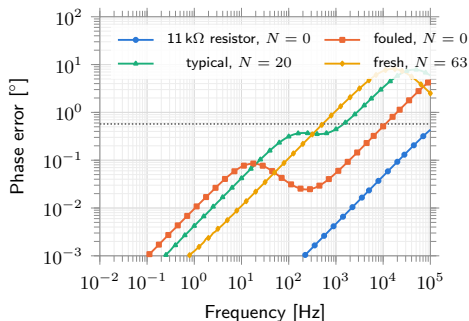
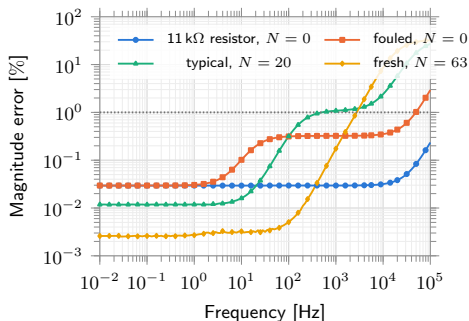
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From currents to a scope

References

Backup



Excitation	Demodulation	Accuracy	$ Z $ range
2-code DAC square	firmware, channel hart	$\pm 1\%$, $\pm 0.6^\circ$ to 5.05 kHz; $\pm 5\%$ to 26.5 kHz	246 Ω – 12.1 M Ω

■ schematic sim, pre-layout

No EIS engine. Impedance is a ratio of two readings through one converter; the DAC drops out of the accuracy budget.

One AFE channel, as simulated

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From currents to a scalar

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Parameter	Value	Condition
CE compliance	13.2 mV – 2.480 V	2.5 V rail, corners
Cell current delivered	$\pm 52.5 \mu\text{A}$	12 k Ω cell
TIA compliance	$\pm 0.80 \text{ V}$ about v_{cm}	
Feedback resistor	19.6 k Ω – 1.93 M Ω , 320 codes	monotonic at every corner
Full-scale current	$\pm 40.7 \mu\text{A}$ ($N=0$) to $\pm 0.41 \mu\text{A}$ ($N=319$)	SAR window $\cap$ TIA compliance
Current quantum	0.824 nA at maximum gain	below the 2 nA noise budget
Bipolar accuracy	0.25–0.38 LSB worst	560 k Ω , 1 M Ω electrode, corners
Control-loop PM	79–80°	five corners, into a cell
TIA-loop PM	89–93°	12-cell electrode grid
Input current noise	341 pA standalone, 3.6–21 nA into an electrode	0.1–100 Hz, 560 k Ω
Zero-current offset	$\sigma(v_{\text{re}}) = 5.72 \text{ mV}$, gain-independent	200 MC samples, calibratable
Quiescent current	197–250 μA per channel	amplifiers + local bias

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

Pre-layout, electrode held nominal: no row includes electrode-to-electrode spread.

The fusion runs on the NPU

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$$s = \sigma\left(\sum_{k=1}^4 w_k \hat{c}_k + b\right)$$

Input	peak current, peak potential, baseline $\times 4$
Mode	0, dense layer, one sigmoid
Output	one risk score
Weights	trained off-chip, staged by firmware

Per site

Hart	1, ownership enforced in hardware
Potential	own 12-bit DAC
Converter	own 10-bit SAR
Power	own MTCMOS domain

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

References

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The drip

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- [1] NCI, *SEER Cancer Stat Facts*, per-site pages, seer.cancer.gov/statfacts, accessed 20 Sep. 2026. Stage, survival: SEER 21, 2016–22; cases, deaths: 2026 est.
- [2] D. C. Grossman *et al.*, “Screening for ovarian cancer: USPSTF recommendation statement,” *JAMA* 319:588, 2018.
- [3] USPSTF, *A and B Recommendations; Prostate Cancer: Screening*, 2018; *Pancreatic Cancer: Screening*, 2019. uspreventiveservicestaskforce.org.
- [4] U. Menon *et al.*, “Ovarian cancer population screening and mortality after long-term follow-up in UKCTOCS,” *Lancet* 397:2182, 2021.
- [5] F. Chen *et al.*, “Clinical analysis of four serum tumor markers in 458 patients with ovarian tumors: combined use of HE4, CA125, CA19-9, and CEA,” *Cancer Manag. Res.* 10:1313, 2018.
- [6] R. G. Moore *et al.*, “A novel multiple marker bioassay utilizing HE4 and CA125 for the prediction of ovarian cancer in patients with a pelvic mass,” *Gynecol. Oncol.* 112:40, 2009.
- [7] USPSTF, *Grade Definitions* (current, after July 2012), uspreventiveservicestaskforce.org, accessed 20 Sep. 2026.
- [8] P. Charkhchi *et al.*, “CA125 and ovarian cancer: a comprehensive review,” *Cancers* 12:3730, 2020.

Backup: USPSTF screening grades

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From currents to
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References

Backup

Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	prostate PSA 55–69 (2018)
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.